University at Buffalo

Project 6

Curve fitting using neural network with PB algorithm

In this computer project, we will use a neural network for curve fitting.

1. Draw $n = 300$ real numbers uniformly at random on $[0, 1]$, call them $x_1, \ldots, x_n$.

2. Draw n real numbers uniformly at random on $\left[-\frac{1}{10}, \frac{1}{10}\right]$, call them $\nu_1, \ldots, \nu_n$.

3. Let
$$d_i = \sin(20x_i) + 3x_i + \nu_i, \qquad i = 1, \ldots, n.$$
Plot the points (x_i, d_i), $i = 1, \ldots, n$.

   We will consider a $1 \times N \times 1$ neural network with one input, $N = 24$ hidden neurons, and 1 output neuron. The network will thus have $3N + 1$ weights including biases. Let w denote the vector of all these $3N + 1$ weights. The output neuron will use the activation function $\phi(v) = v$; all other neurons will use the activation function $\phi(v) = \tanh v$. Given input x, we use the notation $f(x, w)$ to represent the network output.

4. Use the backpropagation algorithm with online learning (update the weights after presenting each input) to find the optimal weights/network that minimize the mean-squared error (MSE)
$$\frac{1}{n} \sum_{i=1}^{n} \left(d_i - f(x_i, w)\right)^2 .$$
Use some η of your choice. Plot the number of epochs vs. the MSE in the backpropagation algorithm.

   Hint: As discussed in class, for a given fixed η, the algorithm may not always result in a monotonically decreasing MSE (the descent may overshoot the locally optimal point). You may have to modify the gradient descent algorithm in such a way that you decrease η (e.g., via $\eta \leftarrow 0.9\eta$) whenever you detect that the MSE has increased. Also, beginning with a very large η may result in an immediate divergence of the weights.

5. Let us call the weights resulting from the backpropagation algorithm (when it converges) as w_0. The curve
$$\big(x, f(x, w_0)\big), \qquad x \in [0, 1]$$
will then be a fit to the points (x_i, d_i), $i = 1, \ldots, n$. Plot the curve $f(x, w_0)$ as x ranges from 0 to 1 on top of the plot of points in (3). The fit should be a "good" fit.

6. Submissions:

   (a) Name your code **Project6** and submit your well commented code.

   (b) Simple report in which write the value of the learning rate(s) and how did you change it to improve the performance (if you did). Also, sketch all the required plots in the handout and put them in the report.